



Process mining: software comparison, trends, and challenges

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Abstract

Process mining is the confluence between data mining and business process management, which is a growing and promising research topic. From process execution event logs, process mining focuses on understanding end-to-end processes and helps provide more significant findings. In this paper, a brief review of each of the main stages (discovery, conformance, and enhancement) of the process mining and low-code automation platforms for business processes are stated. Also, it provides an analysis of the 16 most prominent process mining software as well as an in-depth taxonomy considering 55 features. From this comparison, a subset of software obtained the best scores for process discovery while others for process simulation. Finally, trends and a set of challenges for process mining are pointed out.

Keywords Process mining · Data mining · Software · Process automation

1 Introduction

Nowadays, there are internal processes in companies, which were implemented years ago. These processes have been ongoing from their inception, but process management and improvement have been a back-burner issue for several years due to the complexity of analyzing each end-to-end process [39]. As a consequence, process mining (PM) is becoming a vital tool for the board of directors of several companies, helping them discover and improve the end-to-end process while revitalizing process management in companies [13,28,32,41]. Although PM emerged several years ago [2], it became a hot area a decade ago due to the advances in data mining and machine learning applied to PM with the aim of discovering and improving end-to-end processes [13,35,41].

In business process reengineering, companies are focused on analyzing future processes; consequently, they are limited to exploring how their processes are currently performed. However, understanding all implemented processes is critical to know how many activities affect the process management, if processes mutually affect each other, or if it is worth investing in improvements [34]. As a result, the primary trend of companies is to either avoid current processes analysis internally or not pay for process mining consultants to analyze them [29].

A general problem with process management is that companies have the false premise that using enterprise information systems ensures connections with business processes because they are process-oriented systems. Several of the implemented enterprise systems, such as SAP, Net Suite, Oracle E-Business, and COINCAP, are process-oriented but in the manner that they support processes like order-to-cash [4]. However, it is complicated to understand how these processes are executed from the information system, where the main bottleneck is, or which activities affect end-to-end throughput time [27,37].

Process mining software can help companies' steering and experts easily capture detailed information from their processes and provide about how critical processes are performing. For implementing a complete process mining procedure in companies, there are two main stages: first, business process automation and, after that, discovering, conformance, and improving the automated business processes [13,29,41].

Business process automation (BPA) focuses on restructuring work resources, integrating diverse applications, and using software throughout all processes of the companies. PM is fed on from all event logs generated by BPA to discover and improve these business processes. However, by selecting a prominent PM software, knowing the PM's trends and current challenges in this topic is vital for providing an excellent strategy to companies' steering committees and experts to discover and improve their business processes [13,40,41].

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Hence, this paper will briefly introduce BPA and PM in Sect. 2, emphasizing the PM stages (discovery, conformance, and enhancement). Also, an in-depth comparative analysis of the most prominent state-of-the-art process mining software is provided in Sect. 3. Next, current process mining trends, as well as main challenges, are exposed in Sect. 4. Finally, conclusions and future works are provided in Sect. 5.

2 Related work

This section provides a brief introduction and review of business process automation (Sect. 2.1), process discovery (Sect. 2.2), process conformance (Sect. 2.3), and process enhancement (Sect. 2.4).

2.1 Business process automation

Usually business processes involve many documents, email chains, and delegations of tasks, which are prone to errors if these processes are not automated. Therefore, even the most minor error can lead to inefficiency problems, communication failures, and missed deadlines, which gradually increase. Business process automation (BPA), also known as business automation, focuses on restructuring work resources, integrating different and diverse applications, and using software throughout all processes of the companies [13,41].

Robotic process automation (RPA) is a novel area within BPA, which consists of automating specific repetitive tasks (a.k.a agents or robots) using software prepared to reproduce the decisions that people make (e.g., send emails by using a template) [17]. RPA aims to automate repetitive tasks by using software agents while people can focus on more productive tasks [3]. BPA solutions increasingly use both artificial intelligence (AI) and RPA to interpret unstructured data better, which offers an optimized user experience, as the technology manages to resolve ambiguity and interact with user needs without having to turn to a technical specialist [1].

Nowadays, BPA and RPA can be easily implemented in companies due to low-code automation software. Low-code automation is for automating business processes requiring few or no coding, making it a good option for non-software developers and decreasing reliance on developers, commonly associated with specific programming languages. The main idea is to develop easy and visual tools, such as application design with drag-and-drop capabilities, instead of implementing complex code using specific and high-level programming languages [30,31].

One of the most prominent low-code automation software for business processes is Appian,¹ which has been implemented in several companies, such as Santander, Bayer,

Univision, Pandora, and T-Mobile. Appian aims at a platform as a service (PaaS) for building enterprise software applications with easy user interface and pre-built development modules. Appian allows automating repetitive tasks in an integrated workflow with Appian low-code RPA or third-party RPA. Also, Appian provides integration with Google, AWS, and Azure for processing massive volumes of forms effortlessly by using Artificial Intelligence. As was stated by Appian's web page,² in 2021, it was named Customers' Choice for Enterprise Low-Code Application Platforms and was the only vendor to receive this distinction for Global Enterprises, Large Enterprises, North America, and the Finance Industry.

Kofax offers the Kofax TotalAgility³ product, a powerful all-in-one low-code solution using the industry's leading capture, optical character recognition (OCR), and process orchestration technology by combining document and process intelligence. The low-code automation and end-to-end business process handling provided by Kofax TotalAgility are the core for offering a central intelligent automation platform to handle dynamic tasks, trigger automated rules, and deploy on-demand workforce capacity. Kofax TotalAgility is immersed in applying AI-powered RPA by unlocking document intelligence, connecting disparate systems, and orchestrating human and digital workers for executing and automating workflows across high-value business processes. Also, it provides advanced cognitive capture and AI to unstructured data to extract information and unlock data insights automatically.

RESTsec is a low-code platform based on representational state transfer supporting rapid security requirements modeling for enterprise services [42], abiding by the state-of-the-art attribute-based access control (ABAC) authorization scheme. RESTsec also takes advantage of the 2D-MDE architecture and provides an open room for creating advanced access control paradigms by embedding non-intrusive extensions. As a capacity, RESTsec is enabled to migrate data from an existing database, reducing thus the effort required. RESTsec leaves improving room for including a machine learning (ML) module able to identify general errors, possible logic holes, and conflicting rules. Also, it needs to perform automated access control testing of the generated services [42].

Several platforms and software are designed for BPA and RPA, which have different features, and some would be a better option than others considering both companies and BPA features. Consequently, a comparison software for contrasting BPA and RPA can be consulted online.⁴

¹ <http://appian.com>.

² <http://appian.com/resources/resource-center/analyst-reports/gartner-peer-insights-enterprise-low-code.html>.

³ www.kofax.com/products/totalagility.

⁴ www.g2.com/categories/robotic-process-automation-rpa.

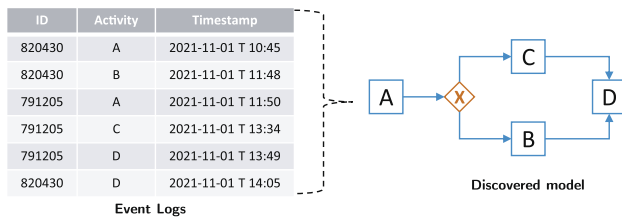


Fig. 1 An example of the discovered model through process discovering by using event logs

2.2 Process discovery

Process discovery (PD) is the most explored approach in PM, where the findings are obtained from an *a posteriori* view of business process execution by using event logs. Usually, these logs are conformed by at least three features: instance (a.k.a case) identifier, activity, and timestamp (see Fig. 1). While the path obtained by going through the activities of a certain instance is known as: trace [6,16,40]. However, other features regarding the business are recommended to be included in event logs to enrich the knowledge that the process miner would extract. PD algorithms should deliver high-quality and comprehensible models by avoiding unnecessary complexity and delivering acceptable accuracy while balancing three main metrics: recall, precision, and generalization [10,13].

PD aims to easily answer three-key questions such as: (i) how are the instances being executed? (ii) what are the highest recurrent paths? and (iii) how is the frequency distribution according to each processing path? Obviously, other more complex questions can be answered, such as (iv) what are the main process's bottlenecks? (v) how is the throughput time affected by the process's main violations? and (vi) what are those patterns describing the best combination of activities for obtaining the shorter and best process's path? [13]. There are several algorithms in the PD approach, which can be grouped as follow:

Deterministic mining- or abstraction-based The most prominent algorithms in this category are Alpha, Beta and Alpha Plus algorithms. They are based on the activities' timestamp, where they find a set of dependency relationships, including causal, parallel, and selection relationships, mapping them to a Petri net model. However, some previous papers reported that their main disadvantage is their sensibility to noise data [13,14,22].

After commenting the main algorithms and the core of mining- or abstraction-based algorithms, the α algorithm can be described as:

1. $T_L = \{t \in T | \exists \sigma \in L, t \in \sigma\}$,
2. $T_I = \{t \in T | \exists \sigma \in L, t = \text{first}(\sigma)\}$,
3. $T_O = \{t \in T | \exists \sigma \in L, t = \text{last}(\sigma)\}$,

4. $X_L = \{(A, B) | A \subseteq T_L \wedge A \neq \emptyset \wedge A \subseteq T_L \wedge B \neq \emptyset \wedge \forall a \in A \forall b \in B \rightarrow L_b \wedge \forall a1, a2 \in A, a1 \# L^{a2} \wedge \forall b1, b2 \in B, b1 \# L^{b2}\}$,
5. $Y_L = \{(A, B) \in X_L | \forall (A', B') \in X_L, A \subseteq A' \wedge B \subseteq B' \implies (A, B) = (A', B')\}$,
6. $P_L = \{p_{(A,B)} | (A, B) \in Y_L\} \cup \{i_L, o_L\}$,
7. $F_L = \{(a, p_{(A,B)}) | (A, B) \in Y_L \wedge a \in A\} \cup \{(p_{(A,B)}, b) | (A, B) \in Y_L \wedge b \in B\} \cup \{(i_L, t) | t \in T_I\} \cup \{(t, O_L) | t \in T_O\}$, and
8. $\alpha(L) = (P_L, T_L, F_L)$

where L is an event log and T is a set of activities. A is the set of input transitions ($\bullet p(A, B) = A$) and B is the set of output transitions ($p(A, B) \bullet = B$).

Heuristic mining or initiative-based They use dependency relationships similar to the deterministic-based, but instead, they have the premise that by increasing the frequency of dependency, the probability of randomness decreases and vice versa [11]. They can address noise and incomplete information, unlike those deterministic-based. Their strategy consists of three main steps: (i) creating a dependency/frequency table from the event log; (ii) creating a dependency/frequency graph from the associated table based on a set of heuristic rules; and (iii) creating a Petri net using the information in the dependency/frequency graph and its table. Heuristics Miner is one of the most prominent heuristic mining or initiative-based algorithms [8,14].

Inductive-based They are based on the divide and conquer mechanism where first, a stochastic activity (SA) directed graph is created for process instances, which include direct dependencies between activities. After that, the resulting SA graph is converted into a workflow model. The main advantage of inductive-based algorithms is their ability to detect duplicate tasks. The most prominent algorithms in this group are mergeSeq, splitSeq, and splitPar [14].

Evolutionary or search-based They are based on the genetic approach; as a consequence, they impose high computational complexity. A genetic-based approach for process mining usually works by initializing a random population of the process model. After the random population is obtained, the fitness index should be computed for each process model forming the partition of the primary population. After that, the next generation is created by crossing over and mutating the initial population. Until a fit model with a high fitness index is found, each generation is crossed over and mutated from an old generation to a new generation. Some of the most prominent evolutionary or search-based algorithms are: NSGAI, NSGA3, hybrid genetic algorithm, and genetic mining [14].

Clustering-based They are primarily based on creating a process model for each instance, which are grouped by using a given traditional clustering-based algorithm. The final model is obtained from these obtained groups. As these

solutions are clustering-based, each instance must be mapped to a vector according to the clustering-based algorithms to be executed. Spaghetti models have several issues, which have been solved by using clustering-based algorithms, such as k-means and AXOR. However, if the event log contains several processes, it is difficult to understand the model and determine the scope of processes by using this clustering-based approach [14].

Finally, in PD, replay allows one to look at each log trace and provides the logic between activities and transitions for each model's instance. Also, replay offers to simulate all model's instances based on the event log from a discovered process model. In this way, replay allows measuring inconsistencies between model and event logs, being a helpful method for behavior and conformance analysis [13].

2.3 Process conformance

Process conformance (PC) aims to measure the quality of a process model. For assessing the quality of a given process model, there are four main metrics [9,13]:

- *Fitness* aims to look at the discovered model's event log behavior (usually, a value of 0 means the absence of capacity for supporting all event traces, while a value of 1 means perfect fitness).
- *Precision* is responsible for monitoring and assessing a correct and related event logging behavior in the discovery process (e.g., avoiding underfitting concept).
- *Generalization* provides the ability to find the main features of previous events used for model discovery and, based on these main features, accept new similar events (e.g., avoiding overfitting concept).
- *Simplicity* is the ability to provide a model as simple as possible.

PC defines the quality dimensions, how to measure them, and the different analyses of both the conformance checking and compliance monitoring. On the one hand, the conformance check is based on posthumous analysis by using event logs in an offline mode. On the other hand, compliance monitoring is based on run-time event logs [6,9,13].

PC also establishes the main c-level interest related to process conformance with the aim of answering some questions, such as how accurate are the processed cases regarding the standardized and documented process models? What are the critical problems? How frequent are the anomalies? and where are the business rules violated? [13]. As a result, [21] identified 10 PC's functionalities, such as (i) time metrics; (ii) case data; (iii) resource information; (iv) explicit/implicit conditions associated with multiple events allowing different orders; (v) supporting activity lifecycle; (vi) multiple-instance constraints, e.g., time, resources, or

other data; (vii) reactively identifying and resolving compliance violations; (viii) proactively detecting and managing violations; (ix) explaining the root cause; and (x) quantifying noncompliance using metrics.

PC's metrics aim to understand, measure, and balance the process model's quality. These metrics are included in the PD approach to support process mining techniques, such as checking conformance, standardization, or harmonization [13,21].

As in every area, conformance checking leaves room for improving it. There are three main challenges to pay attention to:

- *Big data and real-time* Nowadays, the number of records to save is growing exponentially. The larger the process, the larger the event logs. Also, event logs can even save several features and additional data coming from other sources, generating a large amount of information to be recorded in a short time. Conformance techniques must evolve accordingly to handle this exponential growth of information. As a consequence, several process mining software is going to the cloud to take advantage of the computational power and provide fast implantation online while it tries to provide monitoring in real-time [25].
- *Noise, incompleteness, and unfitness* Noise is a well-known problem in process discovery; therefore, conformance approaches must be noise-aware. A way to appear noises in event logs is by those traces incorrectly recorded (e.g., cases with missing information due to temporary system failure) or traces reflecting exceptional situations not representative of the typical behavior of the process. Unfitness is a conformance dimension itself, but it may influence other dimensions: If the model cannot reproduce the observed behavior, it cannot determine in what state the system was at that moment. In short, unfitness refers to those situations where the behavior observed in the log cannot be reproduced by the model [25].
- *Conformance diagnosis* It is often difficult to grasp the causes of the problems in large or highly complex process models. Consequently, it is important to indicate the location of the problems for an adequate conformance diagnosis. However, the diagnosis approaches cannot be limited to simply locating the possible conformance errors, but they must provide aiding mechanisms to the data scientists to fully understand the root causes of the problems [25].

2.4 Process enhancement

Process enhancement (PE) is focused on improving the process model by using external data, statistics and replay, advantages of ML approaches, and process simulation (PS)

[13,14]. The companies' decision-makers invest in improving process models because of the highly obtained business value and insight provided by the improved process. As was stated by [5], PE can help to answer important questions, such as what is the average processing time for cases? What are the transition and decision probabilities? Which transitions are time-consuming? What are the critical activities or resources? What is activity duration (working time)? And how much waiting time between activities?

Replay analysis is a pioneer and visual method widely used in process enhancement, allowing observation of the process execution. By using replay, the companies' decision-makers can suggest improvements to the process model through important visual patterns [13], (see Fig. 2) such as:

- *Activity size* refers to the number of occurrences, metrics related to costs, or resources usage.
- *Activity color* emphasizes throughput time.
- *Connection width* suggests the significance of each dependency or delay.
- *Connections color* highlights the waiting time for executing the next activity and suggests possible bottlenecks.
- *Value* can represent the number of instances transitioning through each activity or the throughput time.

Machine learning has shown to be an important topic for improving several real-world areas [19,26]. PM has benefited from this topic, which has been widely used in PE. By applying ML approaches, such as Bayesian and Naive Bayes networks, decision trees, recommender systems, neural networks, rule-based mining, and distance-based [8,19,20,24,38]; some important predictions can be executed to help decision-makers. For example, based on historical data, ML models can be trained for interesting predictions, such as predicting instance duration, identifying delays, probability of next activity, and recommending a path of execution inside a process [13].

Some of the most prominent proposals in PE using ML are TILDE based on decision trees, which can predict each delay and the delay range. TILDE generates rules for estimating the impact of each user involved in each activity. Evolutionary algorithms (EA) can improve process performance and mitigate risks in real-time by taking proactive and corrective actions. Several rules are extracted using EA to provide further insight into the business processes. RNN–LSTM is a deep learning proposal using two hidden recurrent neural network (RNN) layers and long short-term memory (LSTM) cells to predict the next business process's activity. For this deep learning proposal, the event logs were converted into a new representation based on words to take advantage of the RNN structure working with this representation [13,18].

Finally, one of the most used techniques in PE is process simulation, which aims to simulate several scenarios of a

given business process. Process simulation can describe processes in flow diagrams where unit operations are positioned and connected to by-product streams. Process simulation uses statistics for simulating different scenarios from an initial business model, where business and process experts test different scenarios by rearranging the activities or adding new activities and connections. By doing that, process simulation increases capacity, decreases costs, and reduces cycle time by using a risk-free environment before committing capital and resources. Several software are designed for process simulation (e.g., Process Simulator⁵ of ProModel), which provide support for both in the cloud and local as well as several features for obtaining excellent and accurate simulations.

To better understand the discovery, conformance, and enhancement stages, assume that the following process is related to a driver's license exam. The activities describing that process are:

- R*—exam enrollment
- T*—theoretical exam
- P*—practical exam
- D*—evaluate result
- A*—register for an additional attempt
- O*—obtain degree

Figure 3 shows the discovery, conformance, and enhancement stages for the aforementioned exam process. On the one hand, discovering starts by using event logs containing the cases. The traces are extracted from them using the identifier, activity, and timestamp features (see Fig. 1). Then, a process model is inferred from the cases and traces in the event logs. For example, notice that in section (a) of Fig. 3, both the traces and the process model start with the activity *R* and end with the activity *O*. In addition, the *T* and/or *P* activities always occur after *R*. As a result, from section (a) of Fig. 3, the traces are:

1. RPDO
2. RTPDAPDO
3. RPTDO
4. RTPDATPDO
5. RTPOD
6. RPDAPOD
7. RPTDAPTDATPDO

On the other hand, the conformance stage starts by checking the provided process model and if all cases in the event logs correspond to the occurrences of activities in the proposed model. From the list of traces, notice that the first one shown cannot be replayed because activity *T* is missing. Also, notice that for fifth and sixth traces cannot be replayed

⁵ www.promodel.com/products/ProcessSimulator.

Table 1 Taxonomy of the most prominent process mining software considering several features (Part I)

Feature\software	ABBY Timeline	Apromore	ARIS	Business optix	Celonis	Disco	Ever flow	Logpickr
<i>Data management</i>								
Deployment	0.50	0.50	0.50	0.25	0.75	0.25	0.50	0.50
Import file types	0.17	0.50	0.33	0.50	0.50	0.67	0.83	0.33
Database connections	0.00	0.00	0.33	0.33	0.33	0.33	0.67	0.67
Adapters/connectors	0.36	0.09	0.18	0.18	0.64	0.09	0.45	0.09
Integrated ETL functionality	1.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Data anonymization	0.00	0.00	1.00	0.00	1.00	1.00	1.00	0.00
Data loading	1.00	0.00	1.00	1.00	1.00	0.00	1.00	0.50
Character encodings	0.33	0.17	1.00	0.17	0.17	0.17	0.33	0.17
Attribute types	0.50	1.00	1.00	1.00	1.00	0.50	1.00	0.50
Specify business hours	0.50	0.00	0.75	0.00	1.00	0.75	0.00	0.00
Define event order	0.50	0.00	0.50	0.00	0.50	0.00	0.50	0.00
Start/end timestamp	0.50	1.00	1.00	1.00	1.00	1.00	1.00	1.00
<i>Process graph</i>								
As-Is process visualization	0.29	0.43	0.29	0.14	0.14	0.14	0.43	0.29
Export As-Is process graph	0.40	0.60	1.00	1.00	0.20	0.80	0.40	0.60
Performance highlighting	0.50	1.00	1.00	0.00	1.00	1.00	0.00	1.00
Process animation (replay)	0.75	0.75	0.00	0.00	0.50	0.75	0.75	0.50
Search and filter in graph	0.00	0.50	0.50	0.50	1.00	1.00	0.00	0.50
Graph abstraction	0.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Frequency metrics	0.20	0.60	0.40	0.20	0.80	0.80	0.60	0.40
Time metrics	0.50	0.50	0.50	0.50	1.00	0.50	0.50	0.50
Additional graph metrics	0.00	0.00	1.00	0.00	1.00	1.00	0.00	0.50
<i>Process analysis & analytics</i>								
Process benchmarking	0.50	1.00	1.00	1.00	1.00	0.00	1.00	0.00
Process benchmarking (different logs)	0.50	1.00	1.00	1.00	1.00	0.00	0.00	0.00
Root cause analysis	0.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
Variant breakdown by	0.80	0.60	0.40	0.20	0.60	0.60	0.20	0.60
Case and activity list	0.50	0.50	0.50	0.00	0.50	0.50	0.50	0.00
View case details	1.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00
Rework analysis	1.00	1.00	0.00	0.00	1.00	1.00	1.00	0.00
Edge/transition details	0.50	0.00	0.00	0.00	1.00	1.00	0.50	0.00
<i>Conformance checking</i>								
Compare As-Is and target process	0.00	1.00	1.00	1.00	1.00	0.00	1.00	0.00
Target model creation	0.00	0.67	0.50	1.00	0.50	0.00	0.33	0.00
In-graph conformance visualization	0.00	1.00	0.00	1.00	1.00	0.00	0.00	0.00
List of conformance violations	0.00	1.00	1.00	1.00	1.00	0.00	1.00	0.00
Four-eyes principle	0.00	1.00	0.00	0.00	1.00	1.00	0.00	0.00
Sequence filtering	1.00	1.00	0.00	0.00	1.00	1.00	0.00	0.00
Conformance root cause analysis	0.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00

ABBY Timeline (www.abby.com), Apromore (<https://apromore.com>), ARIS (<https://aris-process-mining.com>), BusinessOptix (www.businessoptix.com), Celonis (www.celonis.com), Disco (<https://fluxicon.com/disco/>), EverFlow (<https://everflow.ai>), Logpickr (<https://www.logpickr.com>).

Table 2 Taxonomy of the most prominent process mining software considering several features (Part II)

Feature\software	ABBYY timeline	Apro more	ARIS	Business optix	Celonis	Disco	Ever flow	Logpickr
<i>Operational support</i>								
Alert generation	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
Predictive analytics	1.00	1.00	0.00	0.00	1.00	0.00	1.00	0.00
Recommendations (prescriptive analytics)	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00
<i>Advanced enhancement capabilities</i>								
Organizational mining	0.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00
Scenario simulation	0.00	0.00	0.00	1.00	1.00	0.00	1.00	0.00
Decision rule mining	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00
<i>Views, monitoring and reporting</i>								
Export reports	0.33	0.11	0.89	0.00	0.56	0.89	0.44	0.00
Export charts and tables	0.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00
Custom dashboards	0.50	0.50	0.50	1.00	1.00	0.00	0.50	0.00
Custom metrics/KPIs	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
KPI Threshold	1.00	1.00	1.00	0.00	1.00	0.00	0.00	0.00
Advanced charts	1.00	1.00	1.00	0.00	1.00	0.00	0.00	0.00
World map	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00
Save filter settings	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00
UI languages	0.17	0.06	1.00	0.22	0.33	0.06	0.11	0.06
Share & collaborate	0.14	0.14	0.29	0.57	0.29	0.29	0.14	0.00
<i>Security & compliance</i>								
Role-based access	1.00	1.00	1.00	1.00	1.00	0.00	1.00	0.00
User authentication	0.67	0.33	0.50	0.33	1.00	0.17	0.17	0.17
Audit logs	1.00	1.00	1.00	1.00	1.00	0.00	1.00	0.00
Total	23.61	26.55	34.86	21.10	45.30	21.25	30.86	9.87

ABBYY Timeline (www.abbyy.com), Apromore (<https://apromore.com>), ARIS (<https://aris-process-mining.com>), BusinessOptix (www.businessoptix.com), Celonis (www.celonis.com), Disco (<https://fluxicon.com/disco/>), EverFlow (<https://everflow.ai>), Logpickr (<https://www.logpickr.com>).

and reporting, and security & compliance. Tables 1, 2, 3, and 4 show binary features where a value of 1 is assigned if the PM software includes the feature and 0 for otherwise. Also, it includes no-binary features where a value, ranging from 0 to 1, is computed according to the number of values available by the software between the total number of all possible analyzed software options. For example, suppose we are analyzing a feature for three process mining tools, labeled Alpha, Bravo, and Charlie, where for the analyzing feature, Alpha provides *a* and *b* options; Bravo provides *a*, *c*, and *d* options; and Charlie provides *d* and *e* options. Then, the total number of all possible analyzed software options for that feature is five (*a*, *b*, *c*, *d*, *e*). Consequently, the score for Alpha is 2/5; Bravo obtains 3/5; and Charlie obtains 2/5. On the one hand, the features containing binary values are: Integrated ETL Functionality, Data Anonymization, Graph Abstraction, Root Cause Analysis, View Case Details, Rework Analysis, In-Graph Conformance Visualization, List of Conformance Violations, Sequence Filtering, Alert Generation, Predictive Analytics, Recommendations (Prescriptive

Analytics), Export Charts and Tables, Custom Dashboards, Custom Metrics/KPIs, Role-Based Access, and Audit Logs. On the other hand, the features containing no-binary values are stated in Table 5.

From this taxonomy, Celonis achieved the best score (TOTAL) regarding the remaining other compared PM software. Celonis reached a high score for almost every feature, except for decision rule mining. Based on this taxonomy, Celonis is highly recommended for process animation (replay) and conformance checking, even for creating some basic simulation of “what-if” scenarios⁷ but it provides few options for an in-depth process simulation analysis. Also, Celonis does not offer a decision rule mining module; although, it provides an ML module,⁸ which can be used as a Python programming notebook and utilize libraries to extract rules and patterns from scratch. However, EverFlow and myInvenio are good options for replay and process simu-

⁷ www.celonis.com/ems/platform/planning-and-simulation.

⁸ www.celonis.com/intelligent-business-cloud/machine-learning.

Table 3 Taxonomy of the most prominent process mining software considering several features (Part III)

Feature\software	MEHR WERK	Minit	my Invenio	PAFnow	Pro discovery	QPR	Signavio	UiPath
<i>Data management</i>								
Deployment	0.75	0.75	0.50	1.00	0.50	1.00	0.25	0.75
Import file types	0.83	1.00	0.33	0.50	0.33	0.33	0.33	0.83
Database connections	0.33	0.33	1.00	0.33	0.33	0.33	0.00	0.33
Adapters/connectors	0.09	0.36	0.82	0.45	0.09	1.00	0.45	0.09
Integrated ETL functionality	1.00	0.00	0.00	1.00	0.00	1.00	0.00	1.00
Data anonymization	1.00	0.00	0.00	1.00	0.00	0.00	1.00	1.00
Data loading	1.00	0.50	1.00	1.00	0.50	1.00	1.00	1.00
Character encodings	0.17	0.17	0.33	0.17	0.33	0.17	0.17	0.33
Attribute types	1.00	1.00	1.00	1.00	0.50	1.00	1.00	1.00
Specify business hours	0.75	0.75	0.50	1.00	0.50	0.00	0.00	0.00
Define event order	0.50	0.00	0.00	0.50	1.00	1.00	0.00	0.50
Start/end timestamp	1.00	1.00	1.00	1.00	1.00	0.50	1.00	1.00
<i>Process graph</i>								
As-Is process visualization	0.14	0.29	0.43	0.29	0.29	1.00	0.14	0.43
Export As-Is process graph	0.80	0.60	0.80	0.00	0.40	0.20	0.00	0.60
Performance highlighting	1.00	1.00	1.00	1.00	0.50	0.00	0.50	1.00
Process animation (replay)	0.00	0.50	0.75	0.00	0.75	0.00	0.00	1.00
Search and filter in graph	0.50	1.00	1.00	1.00	1.00	0.50	0.50	1.00
Graph abstraction	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Frequency metrics	0.40	1.00	0.40	0.60	0.60	0.40	0.20	1.00
Time metrics	0.50	0.50	0.50	1.00	0.50	1.00	0.50	1.00
Additional graph metrics	1.00	0.00	0.50	0.50	0.00	0.50	0.00	1.00
<i>Process analysis & analytics</i>								
Process benchmarking	1.00	1.00	1.00	1.00	0.00	1.00	0.50	1.00
Process benchmarking (different logs)	1.00	1.00	1.00	1.00	0.00	0.00	0.00	1.00
Root cause analysis	1.00	1.00	0.00	1.00	0.00	1.00	0.00	0.00
Variant breakdown by	0.20	1.00	0.60	0.40	0.20	0.20	0.60	0.40
Case and activity list	0.50	0.50	0.50	0.00	1.00	0.50	0.33	0.50
View case details	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Rework analysis	1.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00
Edge/transition details	1.00	1.00	0.50	1.00	1.00	1.00	0.00	0.50
<i>Conformance checking</i>								
Compare As-Is and target process	1.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00
Target model creation	0.33	0.33	0.67	0.17	0.00	0.67	0.50	0.17
In-graph conformance visualization	1.00	1.00	1.00	1.00	0.00	0.00	1.00	1.00
List of conformance violations	1.00	0.00	1.00	0.00	0.00	1.00	0.00	0.00
Four-eyes principle	0.00	1.00	0.00	1.00	0.00	1.00	0.00	0.00
Sequence filtering	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Conformance root cause analysis	0.00	0.00	1.00	0.00	0.00	1.00	0.00	0.00

MEHRWERK (<https://mpm-processmining.com>), Minit (www.minit.io), myInvenio (www.ibm.com/cloud/cloud-pak-for-business-automation/process-mining), PAFnow (<https://pafnow.com>), ProDiscovery (www.puzzledata.com), QPR (www.qpr.com), Signavio (www.signavio.com), UiPath (www.uipath.com).

Table 4 Taxonomy of the most prominent process mining software considering several features (Part IV)

Feature\software	MEHR WERK	Minit	my Invenio	PAFnow	Pro discovery	QPR	Signavio	UiPath
<i>Operational support</i>								
Alert generation	1.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00
Predictive analytics	1.00	0.00	1.00	1.00	0.00	1.00	0.00	0.00
Recommendations (prescriptive analytics)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Advanced enhancement capabilities</i>								
Organizational mining	0.00	1.00	1.00	0.00	1.00	0.00	0.00	1.00
Scenario simulation	0.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00
Decision rule mining	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
<i>Views, monitoring and reporting</i>								
Export reports	0.78	1.00	0.67	0.33	0.67	0.67	0.00	0.78
Export charts and tables	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00
Custom dashboards	1.00	1.00	1.00	1.00	1.00	1.00	0.50	1.00
Custom metrics/KPIs	0.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00
KPI threshold	1.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00
Advanced charts	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00
World map	1.00	1.00	0.00	1.00	0.00	0.00	0.00	0.00
Save filter settings	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
UI languages	0.17	0.28	0.28	0.06	0.11	0.06	0.17	0.06
Share & collaborate	0.29	0.71	0.43	1.00	0.00	0.00	0.00	0.14
<i>Security & compliance</i>								
Role-based access	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
User authentication	0.50	0.33	0.50	0.33	0.17	0.67	0.83	0.50
Audit logs	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Total	36.53	36.91	38.00	37.63	22.27	34.69	22.48	34.91

MEHRWERK (<https://mpm-processmining.com>), Minit (www.minit.io), myInvenio (www.ibm.com/cloud/cloud-pak-for-business-automation/process-mining), PAFnow (<https://pafnow.com>), ProDiscovery (www.puzzledata.com), QPR (www.qpr.com), Signavio (www.signavio.com), UiPath (www.uipath.com)

lation, including decision rule mining, although they cannot obtain good information on conformance checking.

For better understand, Fig. 4 shows a radar chart showing the score for each set of feature, which were grouped into different categories. The score was computed by averaging the score obtained for each feature, which were categorized as shown in Tables 1, 2, 3 and 4. Notice that Logpicks and Disco have the worst scores in the security & compliance and advanced enhancement capabilities categories while they have obtained lower scores in data management, conformance checking, and views, monitoring and reporting. Conversely, myInvenio, EverFlow, and Celonis have obtained the best scores in the advanced enhancement capabilities category, which includes organizational mining, scenario simulation, and decision rule mining. On the one hand, the most reviewed process mining software provide cloud options and with it a way of interacting from different locations. On the other hand, BusinessOptix and Disco only provide the on-premises option, which limits the interaction with other users and locations. From all analyzed process

mining software, Celonis provides the best trade-off among the analyzed features regarding the compared software. Celonis supports an excellent integration with Appian, as process automation, and it also offers several types of user licenses, adjusting to the user's needs.

4 Trends and challenges

Nowadays, there exists a growth in automatizing business processes and process discovery. One of the PM trends is RPA, where each routine is analyzed to determine if it can be automated. RPA aims to relieve workers from non-value-adding routines and enable them to focus on what matters for the customers. Another trend is “what-if” PM, which aims to understand the effect of one or more changes on a business process. Traditional PM is focused on “as is” analysis questions, such as what are the current bottlenecks in the analyzed process? Where are the current sources of waste? Where are the reworking activities? What are the main vio-

Table 5 Features and their no-binary values for comparing the process mining software

Feature	Values
Deployment	Cloud, Hybrid, On-Premises, Standalone
Import file types	CSV, XES, MXML, XLS(X)(M)(B), MDB, ACCDB
Database connections	Oracle, AS/400, JDBC drivers
Adapters/connectors	Microsoft Dynamics AX & NAV, SAP, Oracle, Salesforce, Epicor, ServiceNow, Infor M3, IFS, CDK AutoMaster, M-Files, generic Web Services
Data loading	Data Refresh, Scheduled Jobs
Character encodings	UTF-8 (verified), UTF-16LE, ISO-8859-1, ISO-8859-15, ASCII, Windows-1252
Attribute types	Case-level, Event-level
Specify business hours	Working week, Multiple shifts/day, Exclude days, Holiday calendar
Define event order	By selected column, by custom sequence of activities
Start/end timestamp	2 timestamps
As-Is process visualization	Directly-Follows Graph (horizontal, vertical, circular), Force-Directed Graph, Radial, Tree, Grid
Export As-Is process graph	JPG, PNG, GIF, SVG, BMP
Performance highlighting	Active time, Idle time
Process animation (replay)	Adjust speed, Adjust timeframe, Switch time mode, Zoom in case
Search and filter in graph	Search, Filter
Frequency metrics	Activity frequency, Case frequency, Max. occurrence in case, Start/end activity frequency, Rework metrics
Time metrics	Avg/med/trim med duration, Customizable
Additional graph metrics	Cost metrics, Custom metrics
Process benchmarking	Visual comparison, Metric comparison
Process benchmarking (different logs)	Visual comparison, Metric comparison
Variant breakdown by	Case count, event count, tot/min/max/avg duration and active/waiting time, min/max/avg utilization, finance metrics
Case and activity list	Case count, event count, unique activity count, case-level performance (duration, active time, waiting time)
Edge/transition details	List of all transitions, From-to activities
Target model creation	Import model(BPMN, XPDL, Nimbus, Bizagi + more), Auto-create from as-is, Create new
Export reports	Events (CSV, XES, MXML, Qlik), Cases (CSV, Qlik), Variants (PNG, BPMN, XML)
Custom dashboards	Custom charts, Custom tables
UI Languages	Danish, German, English, Spanish, French, Italian, Hungarian, Dutch, Polish, Portuguese, Swedish, Turkish, Czech, Russian, Arabic, Chinese, Japanese, Korean
Share & collaborate	Cloud, Hybrid, On-Premises, Standalone
User authentication	Basic, 2FA (email), IP-based access restrictions, SSO via SAML 2.0, LDAP, OIDC

lations regarding an ideal model? However, “what-if” PM is focused on data-driven process simulation, addressing questions, such as what would happen if the number of customer orders triples next week? What happens today if 25% of our workforce is confined by COVID19 next month? How much will the process costs be reduced if we automate 95% of the cases for an activity? [12,22]

PM is mainly based on visualization and dashboarding methods, requiring an expert to navigate through them to identify issues in the process and how to address these issues. These methods can indicate to experts if there are bottlenecks or rework loops, but they cannot indicate how to address these issues; for example, should we reallocate resources? Should some activities be automated, and which ones? Should be changed the way an activity be performed? Should it be done

an activity earlier or rather postponed to later in the process? A current trend is automated process improvement, where several companies are applying it to automatically explore a space of potential business process changes and discover combinations of changes to optimize one or more process performance indicators [15,23].

PM has been little studied along with the latest advances in machine learning, which leaves a challenge room for future research. One opportunity for improving PM is to include new ML approaches for providing explainability to experts in a language close to them [19,20,24]. A novel approach is to use contrast pattern mining for extracting the knowledge about those subsets of activities and external data describing the throughput time. For example, $[\text{reworks} \in [10, 15]] \wedge [\text{bottlenecks} > 10] \wedge [\text{wire_transfer_type} = \text{“SWIFT”}]$ is a

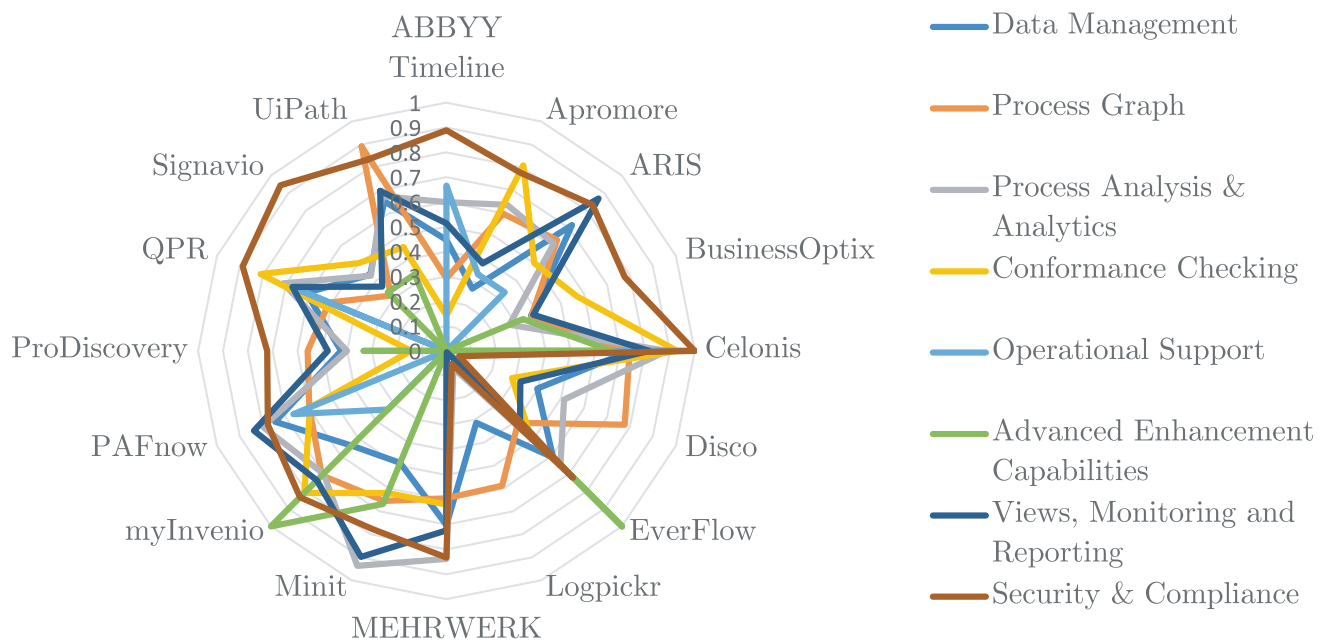


Fig. 4 A radar chart showing the score for each set of features, which are contained in Tables 1, 2, 3 and 4

pattern describing a collection of bank transfers with a high throughput time. One challenge room in PM is to use the reinforcement learning approach for simulating several scenarios and interactions among activities with the aim of optimizing a given process. By using reinforcement learning, agents can determine which activities can be removed or added even change their order. In this way, new scenarios and an optimized process model can be discovered by using the power of the reinforcement learning approach [36]. Data extraction is one of the most challenging because poor data extraction can provide an inaccurate mapping of business processes. On the one hand, several companies do not have event logs for each end-to-end process because some devices interacting in the process do not capture logs. Although some PM software are providing tools for capturing almost all tasks on these devices (e.g., Task mining of Celonis) or some approximate data can be provided by using statistics, it continues to be an open problem. On the other hand, extracted data could contain missing values, which should be used by the process discovery approach without modifying or filling these missing values. Finally, another challenge is to include the PM in the company's culture; in this way, companies need to make sure that they establish their PM expertise in the whole process operation area. Also, it is essential that PM acquired knowledge from people inside the company or hired external consultants to be transferred to the company's steering committee and all employees working in the analyzed process [7].

5 Conclusions

Process mining is a young field joining the forces of process management and data mining, which has gained significant interest from both research and companies. In this paper, a brief review of each of the main stages of process mining and low-code automation platforms for business processes was stated. Also, an analysis of the 16 most prominent process mining software and an in-depth taxonomy considering 55 features was provided. From this review and the process mining software analysis, Appian provided the best option for process automation as a low-code automation software. On the one hand, myInvenio, EverFlow, and Celonis have obtained the best scores in the advanced enhancement capabilities category, which includes: organizational mining, scenario simulation, and decision rule mining. It is important to highlight that Celonis has scored the best out of the remaining benchmarked 15 process mining software considering 55 features. On the other hand, Logpickr and Disco have the worst scores in the security & compliance and advanced enhancement capabilities categories, while they have obtained lower scores in data management, conformance checking, and views, monitoring and reporting. Finally, trends and a set of challenges for process mining were pointed out, where two keystone factors were stated. First, the inclusion of the “what-if” process mining methodology for simulating and predicting possible future scenarios, and preparing a set of actions to avoid possible issues. Second,

contrast pattern mining is a good area to be included to extract the best set of pairs feature-value describing a given target feature. Another opportunity area is to use reinforcement learning for process simulation to execute several scenarios and agents to find the optimal end-to-end process.

As future work, a comparative analysis of the top-ranked process mining software analyzed in this paper but using process mining experts will be executed. The main idea is to evaluate the 55 features analyzed in this paper and a set of other features, such as user interface and user experience, by using an expert-based methodology like the Delphi method. From this analysis, a set of advances and weaknesses can be found, providing help to process mining experts using these process mining software.

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